

NODE Flex

8-Channel Addressable LED Controller · Universal Module Carrier

Wateefy Electronics

One board. Whatever module is in the drawer.

At a Glance

- **8 independent output channels** — WS281x / SK6812 / TM1814 family, 5 V level-shifted drive
- **Any ESP32 DevKit on 0.9" or 1.0" row spacing** — both spacings on the board, up to 22 positions per row, jumper-mapped to the outputs
- **5-48 VDC input, no voltage jumper** — nothing to set wrong, ever
- **30 A through the board at standard copper** — not a paid upgrade
- **Three isolated power domains** — dirty input, clean protected power, fused distribution bus
- **A fuse on every single output** — eight channels, eight fuses, no sharing
- **Non-destructive reverse-polarity protection** — nothing is consumed when someone wires it backwards
- **Four-port DMX-512 / RS-485 output** — per-port transient protection, one port receive-capable
- 123 × 138 mm · 4-layer · ENIG · conformal coated

Why This Exists

Every ESP32 DevKit has a different pinout. Every controller board is hard-wired to exactly one of them. So the board you bought last year is scrap when that module goes end-of-life, and the four dev boards in your parts drawer are useless because none of them match the footprint someone else chose.

NODE Flex breaks that dependency by defining compatibility as **geometry rather than a part number**. The board carries both of the common DevKit row spacings — 0.9 inch and 1.0 inch — with up to 22 positions in each row. Module length does not matter: a 2×15 or 2×19 board seats in the same sockets and simply leaves positions empty. A jumper field then decides which module pin drives which output channel.

The practical result is that the compatibility list cannot go stale. If a module lands on 0.9" or 1.0" centres, it fits — including modules that do not exist yet. When your favourite DevKit is discontinued, and it will be, you seat a different one and keep going. Same board, same protection, same firmware.

For anyone maintaining an installed base across mixed hardware, this is the SKU that ends the inventory problem.

The Architecture: Three Power Domains

Most controllers have one power rail. Input voltage lands on a terminal and it is the same copper that reaches your strips. Whatever arrives — spike, reverse polarity, sag, transient — arrives everywhere at once.

NODE splits power into three separate domains on their own copper planes:

- 1 • **Dirty power in.** The input terminal accepts whatever the supply, the wiring, and the person doing the wiring hand it. This domain is assumed hostile, isolated to the smallest region of the board possible, and nothing downstream touches it directly.
- 2 • **Clean power.** Between dirty and clean sits the protection stage: bidirectional transient clamping and an active reverse-polarity blocker. Reverse the supply and the pass element never turns on — no current path to the rest of the board, nothing sacrificed, red indicator tells you why. Correct the wiring and it recovers on its own.
- 3 • **The protected bus.** Clean power passes the main fuse to become the distribution bus. Each of the eight output channels taps that bus through its own fuse. A fault on run six opens one fuse and the other seven never notice.

BEEFCAKE. The pass element is an 80 V MOSFET whose current rating is several times the board's 30 A design load. It will never be the thing that fails.

Decoupling where the current actually is. Two thousand microfarads of bulk sits on the protected bus, and another forty-seven at every single output connector. The distribution is the point: one reservoir at the input cannot cover the harness inductance of eight separate output runs no matter how large it is. When a full-white frame lands on all eight channels at once, each channel draws from its own reserve sitting centimetres from its terminal.

Four levels between the screw terminal and your pixels — clamp, active block, main fuse, channel fuse — each on its own copper. That is why this is a four-layer board with a split inner power plane rather than two layers and one pour.

Nothing in the pixel power path is rated below **63 V** — and every semiconductor in it is **60 V or above** — against a 48 V absolute maximum input. Nothing in the chain runs near its limit.

At the **36 V nominal** rating, no component in the power path operates above **60 %** of its own rating. At the **48 V absolute maximum**, nothing exceeds **80 %**. On the 5 V, 12 V and 24 V strips that carry the overwhelming majority of installations, the entire plane runs under **40 %**. It is not a number anyone else in this class publishes, and it is the reason to believe the rest of this page.

What Else You Actually Get

Defined-state inputs. Because a channel's jumper may legitimately be left unset, every buffer input carries a pull-down resistor. An unconfigured channel sits at a known logic low instead of floating and driving whatever it feels like — which on a strip full of pixels means darkness rather than noise. This is the detail that separates a real universal carrier from a breakout board with headers.

Data lines treated like data lines. Each output runs through a series resistor and a sub-picofarad ESD device placed on the connector side, where transients actually arrive. The low-capacitance part was chosen deliberately: a fat protection diode on a data line rounds off WS281x bit timing and produces a controller that works on the bench and sparkles on a 40-foot run.

DMX where you need it. Four RS-485 ports on RJ45, each with its own transceiver and its own transient protection, alongside the eight pixel channels.

Three things on this board can be killed by field abuse. All three pull out by hand.

The fuse. A mini blade — the same part sold in a variety pack at every gas station, auto parts counter and big-box store in the country. Pulls with fingers, costs pennies. Not a cartridge you order online and wait for.

The buffer. A jellybean octal logic IC in a DIP socket. It sits directly behind your wiring, which makes it the part most likely to die and the part most

annoying to replace anywhere else. Here it lifts out with a fingernail. And the socket accepts either of two common octal logic families — a strap on pin 1 handles the output-enable behaviour of one and the direction pin of the other — so you fit whichever is in stock. Supply-chain resilience on a fifty-cent line item sounds trivial right up until the next shortage.

The MCU. A socketed module. Blow it up, seat another. No hot air, no braid, no scrap board. Nobody else in this class can claim all three.

And that is what the board area buys. Blade holders, a DIP socket, and terminal clearance cost real estate — there is room to get an iron in, room to pull a fuse without tweezers, room to reach the reset button without unmounting anything, and room at the terminals for heavy wire with ferrules. Compact is easy. Compact *and serviceable* at 30 A is a different problem, and this is the side of it we chose. A channel that dies at eight o'clock on the twenty-third of December should be a fifteen-minute drive, not a three-day wait.

Serviceable by design. The level shifter sits in a DIP socket. The DevKit sits in headers. Fuses pull by hand. Through-hole parts dominate the power section on purpose — this board is meant to be repaired, not replaced.

Where It Goes

- **Mixed-hardware installations** — one controller SKU across a fleet built from whatever modules were available that year
- **Development and evaluation** — swap modules against a known-good, fully protected carrier
- **Long-life deployments** — the board outlives the module generation it was built around
- **Distributed whole-property displays** — one node per zone, eight runs each, scale by adding nodes

How It Compares

Comparison basis. The closest peer is the Bong69 8 Port LED Distro — same idea, power and data integrated on one board, eight ports, WLED, sold at maker scale. QuinLED's Dig-Quad is the other reference point. Competitor figures are taken from the vendors' own published FAQ and specification pages as of July 2026 and should be re-verified before publication.

Athom's WLED range is deliberately excluded: single-strip and PWM controllers aimed at room-level and audio-reactive use. Different product, different buyer.

	NODE Flex	Bong69 8 Port LED Distro	QuinLED-Dig-Quad v3
Output channels	8	8	4
MCU compatibility	Any DevKit on 0.9" or 1.0" row spacing, any length up to 22 positions	Fixed — WT32-ETH01 integrated	Fixed footprint — QuinLED-ESP32 or D1 Mini 32
Survives module obsolescence	Yes — jumper a different DevKit	No — board is the module	No
Input range, one SKU	5–48 VDC	V4: 12–48 V (5 V support dropped) · V3: 5–24 V (discontinued)	5–24 VDC
Covers 5/12/24/36 V pixels	Yes, one board	No — requires choosing a version	Jumper-selectable up to 24 V; 36 V not supported
Input voltage selection	None — nothing to set wrong	Per board version	Jumper; vendor warns wrong setting "will fry the onboard components"
Input connector	Enclosed 57 A screw terminal	V4: spade connectors on a stud	Screw terminal
Continuous current, standard copper	30 A	Not published	15 A at 1 oz (30 A only with paid 2 oz upgrade)
Per-channel fuse rating	7.5 A nominal · 10 A max per output	5 A (V4) · 4 A hold / 8 A trip polyfuse (V3)	10 A max recommended
Fuses on outputs	8 — one per channel	8 — one per channel	5 fuses across 7 outputs (channels share)
Reverse-polarity protection	Active blocking — non-destructive, self-recovering	Not published	Parallel diode with fuse — sacrificial
Input transient clamp	Bidirectional TVS across the input	Not published	Not published
Data-line ESD protection	Per channel, sub-pF device	Series resistor only (33 Ω)	Not published
Power domain separation	3 domains: dirty in / clean / fused bus	Not published	Single rail
Unconfigured input handling	Pull-down on all 8 buffer inputs	n/a — fixed pinout	n/a — fixed pinout
DMX-512 / RS-485 output	4 ports, individually protected	Not published	Not published
Level shifter	8-channel, socketed DIP	Not published	4-channel, soldered
Conformal coating	Yes, applied at the fab	Not offered	Not offered
Fuse-status indicator	Per-channel LED	Blue "good fuse" LED per channel	Per board
Onboard USB programming	Via the module's own port	Onboard USB-C — required, the ESP is soldered down	Via the module's own port
Field-replaceable MCU	Yes — socketed module	No — ESP32-WROOM soldered to the board	Yes — socketed module
Fuse replacement	Mini blade — auto parts store, gas station, variety pack	Cartridge in a surface-mount holder — order online	ATO blade — auto parts store
Audio-reactive / mic support	DIY — headers fitted, spare GPIO available	DIY — wire an INMP441 to the H1 header (documented)	Analog audio input broken out
Temperature sensor support	DIY — headers fitted, spare GPIO available	DIY — add header, DS18B20 and 4.7 kΩ (documented)	Optional onboard footprint
Enclosure	Coming soon	None — 3D-print mounts published	None — bare board
Board size	122 × 128 mm — the cost of blade holders is high	110 × 70 mm	100 × 50 mm

Board size	123 × 138 mm — the cost of blade holders, a DIP socket and terminal clearance	~110 × 70 mm	~100 × 50 mm
Firmware	WLED / ESPHome / ESPixelStick — module-level, not board-level	Same	Same
Firmware image	Pre-built image supplied per module	Pre-built image supplied; ESPixelStick needs the vendor hardware profile	Stock WLED builds
Ecosystem	New	Established, reviewed, xLights docs	Large, mature, well documented

A note on firmware comparisons. WLED, ESPHome, and ESPixelStick are properties of the ESP32 silicon, not of the carrier board — any of them will run on any of these products. What actually differs is which pre-built image ships, and how much GPIO the board's own hardware has already spent before your channels get any.

Where the others are the better choice, plainly. Bong69 ships a proven, well-reviewed board with onboard USB-C programming, per-fuse status LEDs, documented audio-reactive and temperature-sensor support, xLights integration notes, published 3D-print mounts, and boards that arrive pre-configured and tested. QuinLED brings a deep peripheral breakout and years of community documentation. Neither ecosystem exists here yet. If you want established products, buy theirs.

Where NODE Flex is the better choice. Every competing board *is* its module — when that ESP32 variant goes end-of-life, the board goes with it. NODE Flex is a carrier: the module is a consumable, the protected power section is the product. Add one board covering 5/12/24/36 V pixels with every part in the power path well below its rating — no version to pick, eight fuses for eight channels, an enclosed 57 A terminal instead of exposed lugs, and four levels of protection on separate copper.

The honest summary: buy Bong69 or QuinLED if you want the mature ecosystem and never intend to change modules. Buy Flex if you have four different DevKits in a drawer, or if you expect to still be maintaining this installation after the current module generation is gone.

Mounting & Enclosure

No enclosure is included, and none is planned. This is the norm in this segment — bare board, mounted in whatever the installation already uses: a CG1500, a DIN rail, a weatherproof box, a plywood backer.

What NODE adds instead is **conformal coating, applied at the fabrication facility** with connectors, sockets, fuse clips, and switches masked. Competing boards in this class ship uncoated. A coated board in a vented enclosure tolerates condensation and humidity in a way a bare board does not, which matters for the seasonal outdoor installs this was built for.

Four corner mounting holes are provided. *Hole spacing and a downloadable mount STL: to be published at first article.*

Build & Assurance

- **Fabrication** — 4-layer, 1.6 mm, high-T_g substrate, ENIG finish, epoxy-filled and capped vias, IPC Class 2 workmanship
- **Assembly** — professional SMT assembly, flying-probe electrical test on 100% of boards
- **Traceability** — 2D barcode serial on every board; UL-recognized substrate marking
- **Coating** — conformal coating applied at the fab, with connectors, sockets, fuse clips, and switches masked
- **Design review** — every net verified against manufacturer datasheets; schematics independently reviewed by two automated analysis passes with human verification of every finding

Ordering

Item	Detail
Model	NODE Flex
Configuration	Assembled, tested, conformal coated
Included	Controller board, level shifter installed, fuses installed
Not included	Enclosure, ESP32 DevKit module, jumper shunts, 3-pin plug connectors, blade fuses beyond those fitted, power supply
Availability	Pre-release. The present build is a validation run — first-article hardware for bring-up, characterisation and field trial.
Price	TBD
Lead time	TBD
Warranty	TBD

NODE Flex — Technical Specification

Document rev: 3.0 · **Board rev:** V1a · **Date:** 2026-09-01

Components are specified by function and rating. Specific manufacturers and part numbers are selected at build time and may change with availability; every substitution is held to the ratings stated here.

1. Power Architecture

Three separate domains, each on its own copper region:

Domain	Boundary	What it carries
Input (unprotected)	Screw terminal → ideal-diode stage	Raw supply. Assumed hostile. Transient clamp sits across this node.
Clean (protected)	Ideal-diode output → main fuse	Reverse-blocked, surge-clamped. Bulk capacitance lives here.
Distribution bus (fused)	Main fuse → 8 channel fuses + logic regulator	Everything downstream taps this bus through its own fuse.

2. Input & Protection

Parameter	Value	Notes
Pixel input voltage range	5 - 48 VDC	36 V nominal, 48 V absolute maximum
Input connector	Enclosed 2-pole screw terminal	57 A / 10 mm ² rated; 30 A design maximum = 53% of rating
Maximum continuous current	30 A	At standard 1 oz outer copper — no copper upgrade required
Main fuse	30 A mini blade, user-replaceable	Holder rated 30 A / 500 V
Reverse-polarity protection	Ideal-diode controller driving a high-side N-channel pass FET	FET held off under reverse input; no current path to load; non-sacrificial and self-recovering
Pass element	80 V N-channel MOSFET, low milliohm R _{DS(on)}	Rated above the input clamp voltage, not below it; current rating several times the 30 A design load
Input transient clamp	Bidirectional TVS, 1500 W, 48 V standoff	Conducts in neither direction at rated input; polarity-independent
Bulk capacitance	2 × 1000 µF, 63 V	On the clean rail, downstream of the ideal diode
Ideal-diode support	100 V charge-pump capacitor, 80 V hold-up capacitor	Per the controller's reference design
Polarity indication	Anti-parallel green/red LED pair	Green = correct, red = reversed; shared limit resistor sized for the full input range

Voltage headroom summary: 36 V nominal, 48 V absolute maximum. Every component in the power path is rated **63 V or above**; every semiconductor is **60 V or above**.

Derating against input voltage. Utilization of each component's own rating. **5 V, 12 V and 24 V are the mainstream pixel voltages** and are shown in bold:

Component	Rating	5 V	12 V	24 V	36 V	48 V
Bulk input capacitors	63 V	8 %	19 %	38 %	57 %	76 %
Channel output caps	80 V	6 %	15 %	30 %	45 %	60 %
Ideal-diode hold-up	80 V	6 %	15 %	30 %	45 %	60 %
HV-plane ceramics	100 V	5 %	12 %	24 %	36 %	48 %
Ideal-diode controller	65 V	8 %	18 %	37 %	55 %	74 %
Pass MOSFET V _{DS}	80 V	6 %	15 %	30 %	45 %	60 %

At the **36 V nominal** rating no part on the plane exceeds 60 % of its own limit. At the **48 V absolute maximum** nothing exceeds 80 %. On the 5 V, 12 V and 24 V strips that carry the overwhelming majority of installations, the entire plane runs under 40 %.

3. 5 V Logic Rail

Parameter	Value	Notes
Regulator	Wide-input synchronous buck, thermally-padded package	Sat on the distribution bus; 12 thermal vias under the pad
Absolute maximum input	80 V class	Well above the 48 V absolute-maximum input
Switching frequency	400 kHz	Set by a fixed timing resistor
Output voltage	5.1 V nominal	Resistive divider against a 1.0 V reference
Output current capability	2 A	Actual load ≈ 0.5 A → 25% utilization
Inductor	15 µH, 3.0 A saturation, 2.5 A RMS	79% of saturation rating at the full 2 A load
Output capacitance	3 × 22 µF ceramic	Per regulator reference design
Input capacitance	4.7 µF + 100 nF ceramic, 100 V X7R, plus 80 V electrolytic bulk	Rated for the full input range with derating headroom
5 V rail clamp	Unidirectional TVS, 6 V standoff	Cathode to the +5 V rail, anode to ground
Minimum on-time margin	2.7× at 36 V input	No frequency foldback anywhere in the input range
Manual reset	Momentary switch to regulator enable	Pull-up to the distribution bus

4. Outputs

Parameter	Value	Notes
Channels	8	Independent, individually fused — no shared fuses
Output connector	3-pole pluggable terminal, per channel	V+ / DATA / GND; 12 A rated → 63% at the 7.5 A fitted fuse, 83% at the 10 A maximum
Per-channel fuse	7.5 A mini blade fitted, 10 A maximum	25% of holder rating at 7.5 A, 33% at 10 A; user-replaceable
Level translation	Octal logic buffer, DIP-20 socket	Accepts either of two common octal families; 3.3 V drive against a 2.0 V V _{IH} = 1.3 V margin
Data series resistance	330 Ω, axial through-hole, per channel	Limits fault current back into the buffer
Data ESD protection	Sub-picofarad ESD device, per channel, connector side	0.4-0.55 pF junction capacitance; RC ≈ 181 ps — no measurable effect on WS281x bit timing
Data drive current	≈ 4 mA per channel	67% of the ±6 mA buffer rating
Buffer supply current	≈ 45 mA total	64% of the 70 mA device maximum
Per-channel decoupling	47 µF electrolytic 80 V + 100 nF ceramic 100 V	Local to each output connector
Channel activity	One LED per channel	Data activity on that channel

indication		
Fuse-status indication	One LED per channel, fed from the post-fuse node	Lit = that channel's fuse is intact and the bus is live
Buffer input pull-downs	10 kΩ to ground, all 8 inputs	Defines logic low when a channel jumper is not fitted; ~330 μA sink at logic high

4b. DMX-512 / RS-485 Output

Parameter	Value	Notes
Ports	4	RJ45, T568B pairing
Transceivers	One RS-485 transceiver per port	Driver enables tied to a shared enable net; transmit data shared across all four
Receive	Port 1 only	The remaining three ports are transmit-only by design
Port protection	TVS array on the connector-side net, plus series resistance per line	Clamp sits ahead of the series limiting, in that order

5. MCU Interface & Firmware

Parameter	Value
Module support	Any ESP32-family DevKit on a supported row spacing (<i>module not included</i>)
Supported row spacings	0.9" (22.86 mm) and 1.0" (25.4 mm) — both provided on the board
Socket rows	2.54 mm pitch, up to 22 positions per row
Module length	Not constrained — shorter modules (2×15, 2×19) seat in the same rows, leaving positions unused
Channel assignment	Jumper header maps module GPIO to each of the 8 buffer inputs
Unconfigured channels	Held low by pull-down — no floating inputs
Buffer enable	Strap-selectable buffer output enable
Firmware	WLED — pre-built image supplied, configured for the fitted module's GPIO map
Supported LED protocols	WS2811, WS2812/B, WS2813, WS2815, WS2816, SK6812/RGBW, TM1814, and other single-wire families supported by WLED
Connectivity	Per installed module — Wi-Fi 2.4 GHz, or wired Ethernet / PoE-Ethernet where the DevKit provides it
Verified Ethernet DevKits	WT32-ETH01, WT32-ETH01-EVO and WT32-ETH02-PLUS all sit on 0.9" (22.86 mm) centres and seat directly, leaving one free socket row per side
Known non-fitting modules	Olimex and LilyGo boards use non-standard row spacings and will not seat directly — they require an adapter or hand-wiring. Check your module's row centres against 0.9" / 1.0" before ordering
Control protocols	Per WLED: E1.31/sACN, Art-Net, DDP, MQTT, HTTP/JSON API. DMX-512 over the onboard RS-485 ports
Additional I/O	Spare module GPIO broken out on fitted headers for I²C and SPI peripherals, plus 3.3 V / 5 V / GND / TX / RX. Availability depends on the fitted DevKit

Integrator note: confirm that the GPIO assigned to a channel is not a boot-strapping pin on your specific module, and is free of flash or PSRAM duties. Pin restrictions vary by module and remain the integrator's responsibility.

6. Physical

Parameter	Value
PCB dimensions	123 × 138 mm
PCB thickness	1.6 mm
Layer count	4 — signal / ground plane / split power plane / signal
Copper weight	1 oz outer, 1 oz inner
Substrate	High-T _g 155 °C
Surface finish	ENIG, 1 μ" gold
Via treatment	Epoxy filled and capped
Soldermask	Blue
Mounting	4 × corner mounting holes, 3 mm
Mounting hole spacing	107.9 × 103.8 mm
Enclosure	Coming soon
Net weight	TBD — measure at first article
Conformal coating	Applied at the fab; connectors, sockets, fuse clips, and switches masked

7. Environmental & Compliance

Parameter	Value
Operating temperature	−20 °C to +60 °C (<i>design target; not qualification tested</i>)
Substrate rating	UL-recognized, JLC-1 marking
RoHS	Compliant
Workmanship	IPC Class 2

Electrical test	Flying-probe, 100% of boards
Traceability	2D barcode serial number, per board

8. Design Notes & Accepted Limitations

Stated plainly rather than buried:

- **Not automotive qualified.** Protection targets transients credible in a residential or light-commercial installation. Not tested to ISO 7637 load-dump profiles; not intended for vehicle use.
- **Worst-case clamp coordination.** At the TVS datasheet maximum clamping voltage under a full rated pulse, the voltage presented to downstream components exceeds their ratings. That condition requires a pulse energy source which does not exist in a mains-fed low-voltage installation, and is accepted as non-credible for the intended deployment. At realistic clamp currents the entire downstream chain retains margin.
- **Over-voltage beyond specification is not survivable.** A supply above the rated input range will conduct the input TVS and open the main fuse. Sacrificial by design — the board is protected, the TVS and fuse are consumables.
- **Module compatibility is defined mechanically, and validated electrically by the integrator.** Any DevKit on 0.9" or 1.0" row spacing will seat. Whether a given GPIO on that module can drive an output channel is a separate question: strapping pins, flash and PSRAM duties, and input-only pins differ by module and must be checked against the module's own documentation before assigning channels. Olimex and LilyGo boards use non-standard row spacings and will not drop straight in — they need an adapter or hand-wiring to reach the socket rows.
- **Load capability.** The 5 V rail powers onboard logic and indicators only. Pixel power comes directly from the distribution bus; total system current is bounded by the 30 A main fuse and the user's supply.

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